



Right-Triangle Ratios

Grades 9-12 • SAT / ACT mathematics foundations

Geometry and Trigonometry • Use sine, cosine, and tangent in a right triangle.

Original focused skill practice for learners preparing for college-entrance mathematics. Includes models, 16 practice tasks, six explain-and-check tasks, a four-item exit ticket, and worked solutions. Not a full practice exam.

PRINT • TEACH • REVISIT



Plan a focused study session

Learning objective

Use sine, cosine, and tangent in a right triangle.

Readiness

Use after learners have studied the relevant algebra, geometry, or data concepts. Grade tags describe a typical preparation audience; readiness depends on prior coursework.

Suggested use

Session 1: 10 minutes on worked examples and 20–30 minutes on practice. Session 2: review errors, then use explain-and-check tasks. Finish with the independent exit ticket. Pause and reteach a prerequisite when the model is not yet understood.

What this resource covers

The named mathematics skill is useful within SAT and ACT preparation. The packet does not cover every exam domain, reproduce an official section, provide standardized timing, or predict a score.

Materials

Pencil, paper, and a calculator for checking when useful. Practice both symbolic reasoning and calculator verification. Follow each exam publisher's current policies for an actual test.

Student directions

Read the entire question, define quantities, show a method, and check the result. For a written explanation, cite the calculation or feature that supports your conclusion.



Two worked examples

Model 1. In a right triangle, relative to angle θ , the opposite leg is 8 and adjacent leg is 15. Find the hypotenuse and exact $\sin \theta$, $\cos \theta$, and $\tan \theta$.

Hypotenuse = $\sqrt{8^2 + 15^2} = 17$. Use opposite/hypotenuse, adjacent/hypotenuse, and opposite/adjacent.

Model 2. In a right triangle, relative to angle θ , the opposite leg is 24 and adjacent leg is 45. Find the hypotenuse and exact $\sin \theta$, $\cos \theta$, and $\tan \theta$.

Hypotenuse = $\sqrt{24^2 + 45^2} = 51$. Use opposite/hypotenuse, adjacent/hypotenuse, and opposite/adjacent.

Pause before looking at a solution: which relationship, definition, or constraint determines the first step?



Practice • Show a complete method

Show your thinking with a drawing, equation, or explanation. Use the space under each question.

1. In a right triangle, relative to angle θ , the opposite leg is 9 and adjacent leg is 12. Find the hypotenuse and exact $\sin \theta$, $\cos \theta$, and $\tan \theta$.

2. In a right triangle, relative to angle θ , the opposite leg is 5 and adjacent leg is 12. Find the hypotenuse and exact $\sin \theta$, $\cos \theta$, and $\tan \theta$.

3. In a right triangle, relative to angle θ , the opposite leg is 9 and adjacent leg is 12. Find the hypotenuse and exact $\sin \theta$, $\cos \theta$, and $\tan \theta$.



Practice • Show a complete method

Show your thinking with a drawing, equation, or explanation. Use the space under each question.

4. In a right triangle, relative to angle θ , the opposite leg is 8 and adjacent leg is 15. Find the hypotenuse and exact $\sin \theta$, $\cos \theta$, and $\tan \theta$.

5. In a right triangle, relative to angle θ , the opposite leg is 3 and adjacent leg is 4. Find the hypotenuse and exact $\sin \theta$, $\cos \theta$, and $\tan \theta$.

6. In a right triangle, relative to angle θ , the opposite leg is 5 and adjacent leg is 12. Find the hypotenuse and exact $\sin \theta$, $\cos \theta$, and $\tan \theta$.



Practice • Show a complete method

Show your thinking with a drawing, equation, or explanation. Use the space under each question.

7. In a right triangle, relative to angle θ , the opposite leg is 16 and adjacent leg is 30. Find the hypotenuse and exact $\sin \theta$, $\cos \theta$, and $\tan \theta$.

8. In a right triangle, relative to angle θ , the opposite leg is 10 and adjacent leg is 24. Find the hypotenuse and exact $\sin \theta$, $\cos \theta$, and $\tan \theta$.

9. In a right triangle, relative to angle θ , the opposite leg is 9 and adjacent leg is 12. Find the hypotenuse and exact $\sin \theta$, $\cos \theta$, and $\tan \theta$.



Practice • Show a complete method

Show your thinking with a drawing, equation, or explanation. Use the space under each question.

10. In a right triangle, relative to angle θ , the opposite leg is 5 and adjacent leg is 12. Find the hypotenuse and exact $\sin \theta$, $\cos \theta$, and $\tan \theta$.

11. In a right triangle, relative to angle θ , the opposite leg is 5 and adjacent leg is 12. Find the hypotenuse and exact $\sin \theta$, $\cos \theta$, and $\tan \theta$.

12. In a right triangle, relative to angle θ , the opposite leg is 24 and adjacent leg is 45. Find the hypotenuse and exact $\sin \theta$, $\cos \theta$, and $\tan \theta$.



Practice • Show a complete method

Show your thinking with a drawing, equation, or explanation. Use the space under each question.

13. In a right triangle, relative to angle θ , the opposite leg is 24 and adjacent leg is 45. Find the hypotenuse and exact $\sin \theta$, $\cos \theta$, and $\tan \theta$.

14. In a right triangle, relative to angle θ , the opposite leg is 3 and adjacent leg is 4. Find the hypotenuse and exact $\sin \theta$, $\cos \theta$, and $\tan \theta$.

15. In a right triangle, relative to angle θ , the opposite leg is 6 and adjacent leg is 8. Find the hypotenuse and exact $\sin \theta$, $\cos \theta$, and $\tan \theta$.



Practice • Show a complete method

Show your thinking with a drawing, equation, or explanation. Use the space under each question.

16. In a right triangle, relative to angle θ , the opposite leg is 9 and adjacent leg is 12. Find the hypotenuse and exact $\sin \theta$, $\cos \theta$, and $\tan \theta$.



Explain and check • Transfer practice

Solve the task, then explain a check or limitation. Compare your reasoning with the worked key only after making your own attempt.

1. In a right triangle, relative to angle θ , the opposite leg is 9 and adjacent leg is 12. Find the hypotenuse and exact $\sin \theta$, $\cos \theta$, and $\tan \theta$.

2. In a right triangle, relative to angle θ , the opposite leg is 6 and adjacent leg is 8. Find the hypotenuse and exact $\sin \theta$, $\cos \theta$, and $\tan \theta$.

Explain and check • Transfer practice

Solve the task, then explain a check or limitation. Compare your reasoning with the worked key only after making your own attempt.

3. In a right triangle, relative to angle θ , the opposite leg is 3 and adjacent leg is 4. Find the hypotenuse and exact $\sin \theta$, $\cos \theta$, and $\tan \theta$.

4. In a right triangle, relative to angle θ , the opposite leg is 16 and adjacent leg is 30. Find the hypotenuse and exact $\sin \theta$, $\cos \theta$, and $\tan \theta$.



Explain and check • Transfer practice

Solve the task, then explain a check or limitation. Compare your reasoning with the worked key only after making your own attempt.

5. In a right triangle, relative to angle θ , the opposite leg is 24 and adjacent leg is 45. Find the hypotenuse and exact $\sin \theta$, $\cos \theta$, and $\tan \theta$.

6. In a right triangle, relative to angle θ , the opposite leg is 5 and adjacent leg is 12. Find the hypotenuse and exact $\sin \theta$, $\cos \theta$, and $\tan \theta$.

Exit ticket • Independent check

Complete independently. Show reasoning and any relevant restriction. This is a classroom learning check, not a standardized test score.

1. In a right triangle, relative to angle θ , the opposite leg is 15 and adjacent leg is 36. Find the hypotenuse and exact $\sin \theta$, $\cos \theta$, and $\tan \theta$.

2. In a right triangle, relative to angle θ , the opposite leg is 6 and adjacent leg is 8. Find the hypotenuse and exact $\sin \theta$, $\cos \theta$, and $\tan \theta$.



Exit ticket • Independent check

Complete independently. Show reasoning and any relevant restriction. This is a classroom learning check, not a standardized test score.

3. In a right triangle, relative to angle θ , the opposite leg is 8 and adjacent leg is 15. Find the hypotenuse and exact $\sin \theta$, $\cos \theta$, and $\tan \theta$.

4. In a right triangle, relative to angle θ , the opposite leg is 5 and adjacent leg is 12. Find the hypotenuse and exact $\sin \theta$, $\cos \theta$, and $\tan \theta$.



Practice worked solutions

1. 15; $\sin \theta = 3/5$; $\cos \theta = 4/5$; $\tan \theta = 3/4$

Hypotenuse = $\sqrt{9^2 + 12^2} = 15$. Use opposite/hypotenuse, adjacent/hypotenuse, and opposite/adjacent.

2. 13; $\sin \theta = 5/13$; $\cos \theta = 12/13$; $\tan \theta = 5/12$

Hypotenuse = $\sqrt{5^2 + 12^2} = 13$. Use opposite/hypotenuse, adjacent/hypotenuse, and opposite/adjacent.

3. 15; $\sin \theta = 3/5$; $\cos \theta = 4/5$; $\tan \theta = 3/4$

Hypotenuse = $\sqrt{9^2 + 12^2} = 15$. Use opposite/hypotenuse, adjacent/hypotenuse, and opposite/adjacent.

4. 17; $\sin \theta = 8/17$; $\cos \theta = 15/17$; $\tan \theta = 8/15$

Hypotenuse = $\sqrt{8^2 + 15^2} = 17$. Use opposite/hypotenuse, adjacent/hypotenuse, and opposite/adjacent.

Practice worked solutions

5. 5; $\sin \theta = 3/5$; $\cos \theta = 4/5$; $\tan \theta = 3/4$

Hypotenuse = $\sqrt{3^2+4^2} = 5$. Use opposite/hypotenuse, adjacent/hypotenuse, and opposite/adjacent.

6. 13; $\sin \theta = 5/13$; $\cos \theta = 12/13$; $\tan \theta = 5/12$

Hypotenuse = $\sqrt{5^2+12^2} = 13$. Use opposite/hypotenuse, adjacent/hypotenuse, and opposite/adjacent.

7. 34; $\sin \theta = 8/17$; $\cos \theta = 15/17$; $\tan \theta = 8/15$

Hypotenuse = $\sqrt{16^2+30^2} = 34$. Use opposite/hypotenuse, adjacent/hypotenuse, and opposite/adjacent.

8. 26; $\sin \theta = 5/13$; $\cos \theta = 12/13$; $\tan \theta = 5/12$

Hypotenuse = $\sqrt{10^2+24^2} = 26$. Use opposite/hypotenuse, adjacent/hypotenuse, and opposite/adjacent.

Practice worked solutions

9. 15; $\sin \theta = 3/5$; $\cos \theta = 4/5$; $\tan \theta = 3/4$

Hypotenuse = $\sqrt{9^2 + 12^2} = 15$. Use opposite/hypotenuse, adjacent/hypotenuse, and opposite/adjacent.

10. 13; $\sin \theta = 5/13$; $\cos \theta = 12/13$; $\tan \theta = 5/12$

Hypotenuse = $\sqrt{5^2 + 12^2} = 13$. Use opposite/hypotenuse, adjacent/hypotenuse, and opposite/adjacent.

11. 13; $\sin \theta = 5/13$; $\cos \theta = 12/13$; $\tan \theta = 5/12$

Hypotenuse = $\sqrt{5^2 + 12^2} = 13$. Use opposite/hypotenuse, adjacent/hypotenuse, and opposite/adjacent.

12. 51; $\sin \theta = 8/17$; $\cos \theta = 15/17$; $\tan \theta = 8/15$

Hypotenuse = $\sqrt{24^2 + 45^2} = 51$. Use opposite/hypotenuse, adjacent/hypotenuse, and opposite/adjacent.

Practice worked solutions

13. 51; $\sin \theta = 8/17$; $\cos \theta = 15/17$; $\tan \theta = 8/15$

Hypotenuse = $\sqrt{(24^2 + 45^2)} = 51$. Use opposite/hypotenuse, adjacent/hypotenuse, and opposite/adjacent.

14. 5; $\sin \theta = 3/5$; $\cos \theta = 4/5$; $\tan \theta = 3/4$

Hypotenuse = $\sqrt{(3^2 + 4^2)} = 5$. Use opposite/hypotenuse, adjacent/hypotenuse, and opposite/adjacent.

15. 10; $\sin \theta = 3/5$; $\cos \theta = 4/5$; $\tan \theta = 3/4$

Hypotenuse = $\sqrt{(6^2 + 8^2)} = 10$. Use opposite/hypotenuse, adjacent/hypotenuse, and opposite/adjacent.

16. 15; $\sin \theta = 3/5$; $\cos \theta = 4/5$; $\tan \theta = 3/4$

Hypotenuse = $\sqrt{(9^2 + 12^2)} = 15$. Use opposite/hypotenuse, adjacent/hypotenuse, and opposite/adjacent.

Transfer worked solutions

1. 15; $\sin \theta = 3/5$; $\cos \theta = 4/5$; $\tan \theta = 3/4$

Hypotenuse = $\sqrt{9^2 + 12^2} = 15$. Use opposite/hypotenuse, adjacent/hypotenuse, and opposite/adjacent.

2. 10; $\sin \theta = 3/5$; $\cos \theta = 4/5$; $\tan \theta = 3/4$

Hypotenuse = $\sqrt{6^2 + 8^2} = 10$. Use opposite/hypotenuse, adjacent/hypotenuse, and opposite/adjacent.

3. 5; $\sin \theta = 3/5$; $\cos \theta = 4/5$; $\tan \theta = 3/4$

Hypotenuse = $\sqrt{3^2 + 4^2} = 5$. Use opposite/hypotenuse, adjacent/hypotenuse, and opposite/adjacent.

4. 34; $\sin \theta = 8/17$; $\cos \theta = 15/17$; $\tan \theta = 8/15$

Hypotenuse = $\sqrt{16^2 + 30^2} = 34$. Use opposite/hypotenuse, adjacent/hypotenuse, and opposite/adjacent.



Transfer worked solutions

5. 51; $\sin \theta = 8/17$; $\cos \theta = 15/17$; $\tan \theta = 8/15$

Hypotenuse = $\sqrt{(24^2 + 45^2)} = 51$. Use opposite/hypotenuse, adjacent/hypotenuse, and opposite/adjacent.

6. 13; $\sin \theta = 5/13$; $\cos \theta = 12/13$; $\tan \theta = 5/12$

Hypotenuse = $\sqrt{(5^2 + 12^2)} = 13$. Use opposite/hypotenuse, adjacent/hypotenuse, and opposite/adjacent.

Exit-ticket worked solutions

1. 39; $\sin \theta = 5/13$; $\cos \theta = 12/13$; $\tan \theta = 5/12$

Hypotenuse = $\sqrt{15^2 + 36^2} = 39$. Use opposite/hypotenuse, adjacent/hypotenuse, and opposite/adjacent.

2. 10; $\sin \theta = 3/5$; $\cos \theta = 4/5$; $\tan \theta = 3/4$

Hypotenuse = $\sqrt{6^2 + 8^2} = 10$. Use opposite/hypotenuse, adjacent/hypotenuse, and opposite/adjacent.

3. 17; $\sin \theta = 8/17$; $\cos \theta = 15/17$; $\tan \theta = 8/15$

Hypotenuse = $\sqrt{8^2 + 15^2} = 17$. Use opposite/hypotenuse, adjacent/hypotenuse, and opposite/adjacent.

4. 13; $\sin \theta = 5/13$; $\cos \theta = 12/13$; $\tan \theta = 5/12$

Hypotenuse = $\sqrt{5^2 + 12^2} = 13$. Use opposite/hypotenuse, adjacent/hypotenuse, and opposite/adjacent.



Use errors to choose the next step

Identify the first mismatch

Compare the student's representation with the quantities or stated condition. Separate reading the situation, choosing a method, calculation, and interpretation.

Respond precisely

If the model is wrong, use a simpler example before repeating the procedure. If the method is sound and arithmetic is wrong, correct that step and verify the original result.

Review the exit ticket

For each item: 2 points for an accurate result with coherent reasoning; 1 for a sound method with a minor error or an unexplained correct answer; 0 for unsupported or missing reasoning. Use comments to select a next action. These are local instructional criteria.

AIG

Compare two methods, construct a counterexample, or create a new problem with a prescribed solution and justify its restrictions.

504 / EC

Follow the individual plan. Offer manageable item groups, documented time or format supports, a worked step, or an alternate response method while retaining the mathematical objective.

ESL

Preview terms and symbols in the questions. Use home-language rehearsal or a bilingual glossary, then ask for a concise explanation such as "The condition means ___, so I used ___."

Sources, preparation, and license

Objective: Use sine, cosine, and tangent in a right triangle. State alignment: not claimed. This is a focused college-entrance preparation resource, not a state course or a claim of course equivalency.

Official exam scope consulted September 15, 2026: College Board, Types of Math Tested, <https://satsuite.collegeboard.org/sat/whats-on-the-test/math/types> ; ACT, Mathematics Test Description, <https://www.act.org/content/act/en/products-and-services/the-act/test-preparation/description-of-math-test.html> .

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